

AN ISOPARAMETRIC TRANSFORMATION-BASED INTERPOLATION-SUPPLEMENTED LATTICE BOLTZMANN METHOD AND ITS APPLICATION

KUN QU, CHANG SHU^{*} and YONG TIAN CHEW

*Department of Mechanical Engineering, National University of Singapore,
10 Kent Ridge Crescent, Singapore 119260
^{*}mpeshuc@nus.edu.sg*

Received 1 June 2009

Isoparametric transformation was applied to the interpolation-supplemented lattice Boltzmann method (ISLBM). As a result, it can be applied to arbitrarily structural grids and reserve the simple procedure of ISLBM. The simulation of the lid-driven cavity flow and the flow around a cylinder were performed to validate the proposed method. The numerical results agree excellently with the data in the literature.

Keywords: Lattice Boltzmann equation; isoparametric transformation; incompressible flow.

1. Introduction

In recent years, the lattice Boltzmann method (LBM), evolved from the Lattice Gas Automata (LGA),¹ is regarded as an alternative method for simulation of complex flows. However, most of its applications are limited to simple geometries. For the application in complex geometry, there are three ways so far. The first way is solving LBE by FDM,² FVM³ or FEM⁴ The second way is based on the boundary extrapolation on regular mesh. Sometimes they are coupled with mesh refinement.^{5,6} The third way is based on interpolation techniques, such as the interpolation-supplemented LBM (ISLBM),⁷ the generalized form of interpolation supplemented LBM (GILBM)⁸ and the Taylor-series expansion and least-squares-based LBM (TLLBM).^{9,11} These methods reserve the simple streaming-collision procedure of LBM and introduce an interpolation step.

The present approach, isoparametric transformation-based ISLBM (ITB-ISLBM) adopts the isoparametric transformation¹² to implement interpolation on irregular structured grids.

^{*}Corresponding author.

2. Isoparametric Transformation-Based ISLBM (ITB-ISLBM)

Let us write the LBE as the following backward trajectory form

$$f_\alpha(\mathbf{x}, t + \Delta t) = g_\alpha(\mathbf{x} - \mathbf{e}_\alpha \Delta t, t) = f_\alpha(\mathbf{x} - \mathbf{e}_\alpha \Delta t, t) - \frac{f_\alpha(\mathbf{x} - \mathbf{e}_\alpha \Delta t, t) - f_\alpha^{eq}(\mathbf{x} - \mathbf{e}_\alpha \Delta t, t)}{\tau} \quad (1)$$

The set of mesh nodes is $\mathbf{x}$. In order to get $f_\alpha(\mathbf{x}, t + \Delta t)$, $g_\alpha(\mathbf{x} - \mathbf{e}_\alpha \Delta t, t)$ can be interpolated from $g_\alpha(\mathbf{x}, t)$. Since the original LBM is of second order in space, at least we should apply the second order interpolation to reserve the space accuracy.⁷

As shown in Fig. 1, on a structural mesh, D2Q9 model for instance, particles stream from departure points (the solid squares) to a mesh node (the big solid circle). We can select the node and its neighboring 8 nodes to form a quadratic 9-node element. Note that we have to choose a time step small enough to ensure every departure point being in the element. To implement the interpolation step, the element is regarded as a quadrangle transformed from a square in the generalized coordinate system $\xi - \eta$. In the $\xi - \eta$, a smooth function $f(\xi, \eta)$ can be approximated with shape functions

$$f(\xi, \eta) = \sum_{i=1}^9 f_i \phi_i(\xi, \eta) \quad (2)$$

where f_i is the value on the node i , $\phi_i(\xi, \eta)$ is the shape function of the i -th node. For such an isoparametric element, we also have

$$\begin{bmatrix} x & y \end{bmatrix}^T = F(\xi, \eta) = \left[\sum_{i=1}^9 x_i \phi_i(\xi, \eta) \quad \sum_{i=1}^9 y_i \phi_i(\xi, \eta) \right]^T \quad (3)$$

Thus, in order to interpolate $g_\alpha(\mathbf{x} - \mathbf{e}_\alpha \Delta t, t)$ from $g_\alpha(\mathbf{x}, t)$, we need to solve Eq. (3) to get the values of (ξ_p, η_p) of the departure point $\mathbf{x} - \mathbf{e}_\alpha \Delta t$. We adopt the algorithm proposed in Ref. 13 where Newton-Raphson method was used to solve Eq. (3). And the converged solution can be obtained within 4 iterations as long as the point is inside the element. Since the time step Δt is fixed, the generalized coordinates of the departure points are fixed, which can be computed beforehand.

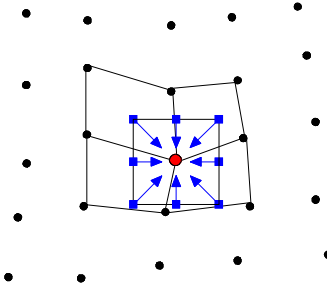


Fig. 1. Schematic plot of backward trajectory.

3. Numerical Tests

In this section, the proposed ITB-ISLBM is validated by simulating the lid-driven cavity flow and flows around a circular cylinder. The D2Q7 lattice velocity model was used in the present work.

3.1. Lid-driven cavity flow

Calculation was performed for $Re = 7500$ on the non-uniform grids of 145×145 . For all the cases, the velocity of the lid is taken as $U = 0.1$, length of the square cavity as $L = 1$, the initial density of the fluid as $\rho = 1$, and the viscosity is determined from $Re = UL/\nu$. Initially, the velocity field was set as zero. The distribution functions along the top wall are set to the equilibrium state.

Table 1. Comparison of the locations of vortices center.

Re		Primary vortex		Lower left vortex		Lower right vortex	
		x	y	x	y	x	y
7500	Ghia ¹⁴	0.5117	0.5322	0.0645	0.1504	0.7813	0.0625
	Hou ¹⁵	0.5176	0.5333	0.0706	0.1529	0.7922	0.0667
	present	0.5121	0.5329	0.0620	0.1581	0.7602	0.0568

Table 1 compares the locations of three vortices obtained by present method with those of Ghia *et al.*¹⁴ and Hou *et al.*¹⁵ Clearly, all the results agree excellently well. As shown in Fig. 2, the profiles of the normalized u on the vertical central line and the normalized v on the horizontal central line are also in good agreement with those of Ghia *et al.*¹⁴

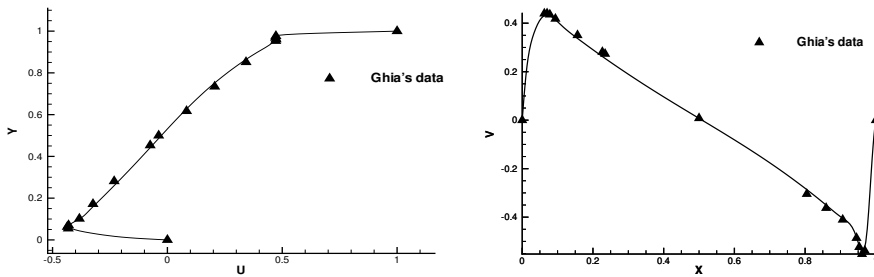


Fig. 2. Normalized U (left) and V (right) velocity profiles along vertical and horizontal central lines with present results (solid lines), as well as the data of Ghia *et al.* (symbols) for $Re = 7500$.

3.2. Flow around an impulsively started circular cylinder

The flows around a circular cylinder of $Re = 10, 20$ and 40 were considered. The Reynolds number is defined as $Re = 2U_\infty r_0 / \nu$, where U_∞ is the free stream velocity and r_0 is the cylinder radius. In this case, we set $U_\infty = 0.05$ and $r_0 = 1$. O-type structured

grids were generated according to Ref. 2. The grid sizes are 91×111 , 91×181 and 181×251 for $Re = 10$, $Re = 20$ and $Re = 40$. As shown in Table 2, the wake length L , separation angle θ_s are in excellent agreements with previous studies.^{2,8,16,19}

Table 2. Comparisons of wake length and separation angle with previous studies.

References	Method	Re = 10		Re = 20		Re = 40	
		L	θ_s	L	θ_s	L	θ_s
Ref. 16	N.S solver	0.53	29.6	1.88	43.7	4.69	53.8
Ref. 17	ISLBM	0.474	26.89	1.84	43.0	4.49	52.8
Ref. 8	GILBM	0.478	26.0	1.85	43.3	4.45	52.4
Ref. 2	FDLBM	0.498	30.0	1.92	42.8	4.51	53.8
Ref. 19	N.S solver	0.434	28.0	1.80	42.1	4.38	50.1
Present	ITB-ISLBM	0.474	29.6	1.79	43.4	4.36	53.3

4. Conclusion

The ITB-ISLBM presented in this work adopts the idea of isoparametric transformation to implement interpolation on irregular structured grids. The numerical experiments on the lid-driven cavity flow and the flow past a circular cylinder show that ITB-ISLBM is a feasible method for problems with irregular geometry.

References

1. U. Frisch, B. Hasslacher and Y. Pomeau, *Phys. Rev. L.* **56** (1986) 1505.
2. R. Mei and W. Shyy, *J. Comput. Ph.* **134**(2) (1997) 306.
3. F. Nanelli and S. Succi, *J. Stat. Phys.* **68** (1992) 401.
4. Y. Li, E. J. Leboeuf and P. K. Basu, *Phys. Rev. E* **69** (2004) 065701.
5. O. Filippova and D. Hänel, *J. Comput. Ph.* **147** (1998) 219.
6. C.-L. Lin and Y. G. Lai, *Phys. Rev. E* **62** (2000) 2219.
7. X. He, L.-S. Luo and M. Dembo, *J. Comput. Ph.* **129** (1996) 357.
8. T. Imamura, K. Suzuki, T. Nakamura and M. Yoshida, *J. Comput. Ph.* **202** (2004) 645.
9. C. Shu, Y. T. Chew and X. D. Niu, *Phys. Rev. E* **64** (2001) 045701.
10. C. Shu, X. D. Niu and Y. T. Chew, *Phys. Rev. E* **65** (2002) 036708.
11. Y. Peng, Y. T. Chew and C. Shu, *Phys. Rev. E* **67** (2003) 026701.
12. R. D. Cook, D. S. Malkus, M. E. Plesha and R. J. Witt, *Concepts and Applications of Finite Element Analysis* (John Wiley and Sons, New York, 2002).
13. A. Allievi and R. Bermejo, *J. Comput. Ph.* **132** (1997) 157.
14. U. Ghia, K. N. Ghia and C. T. Shin, *J. Comput. Ph.* **48** (1982) 387.
15. S. Hou, Q. Zou, S. Chen, G. Dollen and A. C. Cogley, *J. Comput. Ph.* **118** (1995) 329.
16. S. C. R. Dennis and G. Z. Chang, *J. Fluid Mech.* **42** (1970) 471.
17. X. He and G. Doolen, *J. Comput. Ph.* **134** (1997) 306.
18. X. D. Niu, Y. T. Chew and C. Shu, *J. Comput. Ph.* **188** (2003) 176.
19. F. Nieuwstadt and H. B. Keller, *Computer & Fluids* **1** (1973) 59.